

Lecture Summary: 3 points ∇

Over-optimization is a problem... RL in narrow domains is one solution

GRPO is simple (but with some flaws), and enables: **KLVR**

Lots of successful recipes: dpo-RL, kimi 1.5, GPT-4o

MHA/GQA/MLA: head-level attention algorithm

SWA: sliding window attention: token-level attention algorithm

MoE

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GRPO pseudo code ∇

Algorithm 1: iterative group relative policy optimization

Input initial policy model π_{init} ; reward model τ_e ; task prompts $\mathcal{D}$;

hyperparameters ϵ, β, M

$\Rightarrow$ 0: Compute $\bar{A}_i$ for each token of q_i through group relative advantage estimation

$\Rightarrow$ 1: policy model $\pi_0 \leftarrow \pi_{init}$

$\Rightarrow$ 10: for GRPO iteration = 1, ..., U , do:

$\Rightarrow$ 2: for iteration = 1, ..., I do

$\Rightarrow$ 3: Reference model $\pi_{ref} \leftarrow \pi_0$

$\Rightarrow$ 11: update policy model π_0 by maximizing GRPO objective function

$\Rightarrow$ 4: for step = 1, ..., M , do:

$\Rightarrow$ 5:

$\Rightarrow$ 6:

$\Rightarrow$ 7:

$\Rightarrow$ 8:

sample a batch $\mathcal{D}_b$ from $\mathcal{D}$

update the old policy model $\pi_{old} \leftarrow \pi_0$

sample G_i outputs $(O_i)_{i=1}^G \sim \pi_{old}(\cdot | q)$ for each question $q \in \mathcal{D}_b$

compute reward $(r_i)_{i=1}^G$ for each sampled output O_i by τ_e

$\Rightarrow$ output π_0

training using a replay function